

Mini Paper — 2.2 Problem Solving & Programming — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Q1 — [2] (AO1) - A function returns a (single) value (1). - A procedure performs a task but does not return a value (1).

Examiner tip: State the defining difference (returns a value vs does not) — do not write that a function is "more complex".

Q2 — [4] (AO1) - Recursion is a subroutine that calls itself (1). - A correct recursive routine must have a base case that stops the recursion (1). - If the base case is missing or never reached, the routine calls itself indefinitely (1). - Each call adds a frame to the call stack until memory is exhausted — a stack overflow (1). (Max 4)

Examiner tip: Always pair "calls itself" with "base case"; name **stack overflow** for the error.

Q3 — [4] (AO2)

(a) **[3]** - Calls pushed in order: `total(4)` → `total(3)` → `total(2)` → `total(1)` → `total(0)` (base case) (1 for correct order of pushes). - Base case `total(0)` returns 0 (1). - Unwinding (popping): `total(1)=1+0=1`; `total(2)=2+1=3`; `total(3)=3+3=6`; `total(4)=4+6=10` (1 for correct unwinding).

(b) **[1]** - Final value returned = **10** (1).

Examiner tip: Show frames being pushed then popped in LIFO order — do not just compute the answer.

Q4 — [5]

(a) **[2] (AO2)** - First output: **5** (1). - Second output: **15** (1).

(b) **[3] (AO1)** - `changeByValue` passes a copy of the value (1), so the original `num` is unchanged and 5 is printed (1). - `changeByReference` passes the memory address of `num` (1), so the procedure alters the original and 15 is printed (1). (Max 3)

Examiner tip: Examiners want the **consequence** — original unchanged vs original changed — not just "a copy is made".

Q5 — [6] (AO3)

Award up to 6, e.g.: - `class Book` declared with two **private** attributes (1). - Constructor `new` taking parameters and assigning both attributes (1). - `borrow()` reduces `copies` by 1 (1) only when `copies > 0` (selection) (1). - Getter `getCopies()` that **returns** copies (1). - Correct OCR syntax / encapsulation (private attributes, public methods) (1).

```

class Book
  private title
  private copies

  public procedure new(givenTitle, givenCopies)
    title = givenTitle
    copies = givenCopies
  endprocedure

  public procedure borrow()
    if copies > 0 then
      copies = copies - 1
    endif
  endprocedure

  public function getCopies()
    return copies
  endfunction
endclass

```

Examiner tip: Mark **private** on the attributes and a **return** in the getter — both are easy marks candidates often miss.

Q6 — [3] (AO1) - Encapsulation is bundling attributes (data) and the methods that act on them inside a class (1). - Attributes are kept private and accessed only through public methods (1). - Reason: protects data integrity / data hiding / prevents invalid or accidental changes / allows validation in setters (1).

Examiner tip: Keep encapsulation (private data + public methods) distinct from inheritance and polymorphism.

Q7 — [3]

(a) **[1] (AO1) - Data mining** (1).

(b) **[2] (AO2)** - It searches a very large data set (millions of records) to find patterns/correlations not obvious before (1). - Here it reveals which genres are watched together, which informs recommendations (1).

Examiner tip: Always link the method to the scenario detail (large data set, finding hidden patterns) to earn the justification marks.

Q8 — [3] (AO1) - A local variable is declared inside a subroutine and is accessible only within it (1). - A global variable is declared in the main program and is accessible anywhere in the program (1). - Reason to prefer local: fewer side effects / avoids accidental changes from elsewhere / frees memory when the subroutine ends / more modular and easier to debug (1).

Examiner tip: Scope is about **visibility**, not whether the value can change — do not confuse a global with a constant.

Total: 30 marks (Q1 2 + Q2 4 + Q3 4 + Q4 5 + Q5 6 + Q6 3 + Q7 3 + Q8 3).